

Bloch Sphere Geometry — Lecture Slide Deck

Comprehensive visual guide to single-qubit state representations, polar/azimuthal angles, and quantum gate rotations.

1. What is the Bloch Sphere?

The Bloch Sphere is a geometric representation of the state of a single qubit. Each point on the sphere's surface corresponds to a pure quantum state; interior points represent mixed states. The sphere makes superposition, relative phase, and single-qubit rotations visually intuitive.

- **Visual intuition:** Maps complex probability amplitudes to polar and azimuthal angles on a unit sphere for easier reasoning.
- **Gate representation:** Single-qubit unitary quantum gates appear as 3D rotations about specific axes on the sphere.
- **Teaching tool:** Commonly used to introduce qubit behavior, phase shifts, and measurement projection.

2. Sphere Structure — Axes and Poles

The Bloch sphere is a unit sphere (radius $r = 1$) centered at the origin. The canonical axes are defined as:

- **Z-axis:** Computational basis with $|0\rangle$ at the North Pole ($\theta = 0$) and $|1\rangle$ at the South Pole ($\theta = \pi$).
- **X-axis:** Corresponds to the $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ basis.
- **Y-axis:** Corresponds to phase-shifted $|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$ and $|-i\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}$ states.

Points along the equator represent equal-amplitude superpositions differing only by relative phase ϕ .

3. Qubit Parametric Representation

Any pure single-qubit state vector can be written parametrically as:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

- **θ (theta):** Polar angle ($0 \leq \theta \leq \pi$) — controls the relative weights/probabilities between $|0\rangle$ and $|1\rangle$.
- **ϕ (phi):** Azimuthal angle ($0 \leq \phi < 2\pi$) — encodes the relative quantum phase between amplitudes.

■ **Remember:** Global phase is physically irrelevant, but relative phase (ϕ) directly changes quantum interference patterns and measurement outcomes in X/Y bases.

4. Quantum Gates as Rotations

Unitary single-qubit gates correspond to 3D rotations on the Bloch sphere surface:

- **Pauli-X Gate:** π (180°) rotation about the X-axis. Swaps $|0\rangle$ and $|1\rangle$ (bit flip).
- **Pauli-Y Gate:** π rotation about the Y-axis. Bit flip plus phase shift (complex sign changes).
- **Pauli-Z Gate:** π rotation about the Z-axis. Phase flip — leaves computational basis probabilities unchanged.
- **Hadamard (H) Gate:** Rotation that maps poles to the equator, creating equal superposition states and mixing X/Z bases.

5. Measurement & Probabilities

For a general state $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$:

- **Prob($|0\rangle$)** = $|\alpha|^2 = \cos^2(\theta/2)$
- **Prob($|1\rangle$)** = $|\beta|^2 = \sin^2(\theta/2)$

Measurement projects the state to the observed eigenstate; repeated measurements follow the probability distribution set by the polar angle θ .

6. Applications & Limitations

- **Applications:** Essential for single-qubit pulse calibration, control debugging, quantum error mitigation, and interactive visual learning.
- **Limitations:** Directly represents only single-qubit pure states. Mixed states require interior points ($r < 1$) with density matrices. Multi-qubit entanglement cannot be visualized on a single 3D sphere.